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Generative AI governance in higher education: a case study from Africa

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Introduction: The rapid rise of generative artificial intelligence (Gen-AI), particularly large language models (LLMs), is reshaping the higher education landscape. Yet, there is limited empirical documentation of how African universities are integrating Gen-AI into teaching, learning, and research. This study presents a case study of the University of KwaZulu-Natal (UKZN), one of the first African institutions to develop and implement comprehensive academic guidelines for the responsible use of Gen-AI, aligned with national policy priorities and global debates on academic integrity, transparency, and innovation.

Methods: Adopting a qualitative, single-institution case study design, this research draws on process tracing, comparative policy analysis, institutional records, and the authors' direct involvement as members of the AI Task Team. The guideline development process was documented and analysed, from inception and internal deliberation to external peer review, institutional consultation, and final adoption.

Results: The resulting UKZN *AI Academic Guidelines* are based on four foundational principles: encouraging innovation, ensuring ethical and responsible use, maintaining academic rigour, and building institutional capacity. They establish clear policies on Gen-AI adoption across teaching and research, including curriculum integration, standards for disclosure and authorship, approaches to plagiarism, and guidance on data protection. The guidelines also provide a tiered disclosure framework and embed capacity-building initiatives to support AI literacy among staff and students.

Discussion: This case study demonstrates how a higher education institution in the Global South can translate national AI policy into actionable institutional governance while addressing contextual challenges such as resource constraints, digital divides, and multicultural considerations. By framing Gen-AI as an enabling tool rather than a threat, the UKZN model offers a replicable pathway for other African and Global South universities seeking to integrate AI responsibly, enhance academic productivity, and prepare graduates for an AI-driven future.

KEYWORDS

academic integrity, AI literacy, curriculum integration, disclosure framework, generative AI, governance, higher education policy, transparency

1 Introduction

The global rise of generative artificial intelligence (Gen-AI), particularly large language models (LLMs), is reshaping higher education. While many institutions in the Global North have responded with caution, introducing restrictions and compliance-driven policies, there remains limited documentation of how African universities are navigating this pedagogical and technological transformation (Jin et al., 2025; Maina and Kuria, 2024; Sebihi et al., 2025). Continental reviews confirm that while a few institutions in countries such as South Africa, Rwanda, and Nigeria have begun aligning AI policies with national strategies, most African higher education institutions remain in aspirational stages of governance, with uneven readiness and capacity (Sangwa et al., 2025; Sebihi et al., 2025). In particular, some institutional frameworks regard Gen-AI primarily as a problem that must be tightly regulated, permitting its use only under stringent conditions (Rana, 2025; Xiao et al., 2023). This article presents a case study from the University of KwaZulu-Natal (UKZN), a large public research university in South Africa, which adopted a progressive and enabling framework for Gen-AI use in academic contexts. Unlike restrictive models, the UKZN *AI Academic Guidelines* (Amin et al., 2025) explicitly encourage the use of Gen-AI tools to augment academic productivity, stimulate innovation, and facilitate interdisciplinary collaboration. This approach recognises both the transformative potential of Gen-AI and the importance of embedding ethical principles, academic rigour, and contextual sensitivity into institutional practice.

Notably, The UKZN *AI Academic Guidelines* (also referred to as ‘the Guidelines’) were developed in the wake of South Africa’s *National Artificial Intelligence Policy Framework* (Department of Communications and Digital Technologies, 2024) and were significantly shaped by its principles. Unlike many institutional guidelines that predate the national framework, the UKZN Guidelines reflect and operationalise its key priorities, including ethical AI adoption and capacity-building. By aligning with these objectives, the Guidelines not only address the challenges and opportunities of integrating Gen-AI into higher education but also offer a practical pathway for the implementation of the policy framework in academic institutions—for both university staff and students, as called for by Opesemowo & Adekomaya (2024).

1.1 Scholarly discourse on AI tools

Gen-AI tools, such as ChatGPT and other LLMs, have become increasingly integrated into academic workflows, provoking robust discussion about their ethical and practical implications. Central to this discourse are concerns about transparency, the potential stigma associated with Gen-AI use, and the broader risks of misuse or over-reliance on these technologies. Transparency is often highlighted as a cornerstone of responsible Gen-AI use. Calls for disclosure of Gen-AI use in academic writing are grounded in principles of accountability and integrity. For instance, the Russell Group Principles advocate clear policies to ensure that Gen-AI use aligns with academic standards and maintains trust in scholarly work (Russell Group, 2023). Similarly, the

Proposed Harvard AI Code of Conduct emphasises the importance of acknowledging Gen-AI contributions to uphold the credibility of research (Harvard College Students and Teaching Fellows in Creativity et al., 2023). However, the implementation of transparency measures is not without contention. Some argue that mandatory disclosure of Gen-AI use may inadvertently lead to stigma, as highlighted by BaHammam (2023). BaHammam notes that even minor uses of Gen-AI, such as improving grammar and clarity, can invite scepticism from reviewers and readers. This scepticism, rooted in perceived biases against AI-generated contributions, may create barriers to equitable treatment of AI-assisted research. Recent experimental evidence confirms this concern: Cheong et al. (2025) found that both human evaluators and LLM evaluators consistently rated writing less favourably when AI use was disclosed, regardless of content quality. Similarly, others have cautioned that disclosure can trigger heightened scrutiny by editors and peer reviewers (Van Dis et al., 2023), thereby leading to disparities in how AI-assisted manuscripts are evaluated. Survey research across South Africa and beyond echoes these concerns, with academics citing efficiency gains but also raising issues of academic integrity and dependence (Venter et al., 2025). Systematic reviews further highlight recurring risks, including over-reliance and misinformation, which necessitate robust governance mechanisms (Sokhulu et al., 2025). Taken together, these concerns underscore the need for nuanced approaches that balance transparency with the risks of discouraging responsible Gen-AI adoption.

Broader risks associated with Gen-AI adoption have been widely acknowledged. These include over-reliance on these tools, leading to “de-skilling,” which is the potential erosion of human competencies such as critical thinking, nuanced problem-solving, empathy, and ethical judgement (Reinmann, 2023; UNESCO, 2023). In addition, Gen-AI systems can perpetuate biases and misinformation when built on flawed or biased data (UNESCO, 2023). Eke (2023) notably argues that ChatGPT’s ability to create fluent, referenced, and well-structured texts poses risks, such as the outsourcing of academic work, akin to “contract cheating,” especially in the absence of effective tools to detect AI-generated content. While some colleagues share Eke’s concerns about the limitations of current detection tools to detect AI-generated content, they also acknowledge the potential opportunities and benefits that ChatGPT offers for student learning (Sullivan et al., 2023). These benefits include being able to simplify complex concepts, enhancing equity for non-native speakers and students with disabilities through plain language outputs and grammar assistance, and supporting the development of critical thinking skills when used responsibly. Furthermore, when institutions actively encourage the use of Gen-AI tools in academic work, the imperative to implement Gen-AI detection mechanisms may warrant reconsideration. Emphasising detection in such contexts can convey a punitive or mistrustful stance, so potentially conflicting with pedagogical goals that promote innovation, critical engagement, and responsible use of emerging technologies (Cain, 2023). Rather than focusing on detecting Gen-AI use (Balalle and Pannilage, 2025), a more constructive approach could focus on transparency (BaHammam, 2023). Students and academics should be expected to acknowledge any use of Gen-AI tools and clearly articulate the extent and nature of that use. This shift from detection to disclosure supports academic integrity, while aligning with a learning environment that values openness and responsible technological integration (Hosseini et al., 2023; Khalifa and Albadaawy, 2024; Resnik and Hosseini, 2025).

Abbreviations: AI, Artificial Intelligence; Gen-AI, Generative Artificial Intelligence; LLM, Large Language Model; UKZN, University of KwaZulu-Natal; DVC, Deputy Vice-Chancellor.

Viewed through a multicultural lens, others have found that cultural dimensions, such as uncertainty avoidance and long-term orientation, significantly affect perceptions of Gen-AI's ethical and practical implications, while high uncertainty-avoidance cultures are more likely to see use as academic misconduct (Yusuf et al., 2024). The authors explain that cultures with high uncertainty avoidance, which prefer structured rules and predictability, are more likely to perceive the use of Gen-AI as being academic misconduct (Yusuf et al., 2024). These cultures view Gen-AI as a disruptive technology that introduces ambiguity into established norms of academic integrity, such as authorship and accountability. The unpredictability of AI-generated content and the difficulty with detecting its misuse amplify these concerns, thereby leading to a stronger preference for strict regulations or bans to maintain clarity and control in educational practices. This highlights the need for culturally sensitive approaches when integrating Gen-AI into higher education.

The three essential criteria for the ethical use of LLMs in academic writing—human vetting and oversight, substantial human contribution, and clear acknowledgement and transparency—will go a long way in addressing these concerns (Porsdam Mann et al., 2024). This requires teaching students to critically engage with both Gen-AI writing tools and Gen-AI detectors (Wise et al., 2024). These criteria aim to uphold ethical standards, so ensuring originality, fostering responsibility, and promoting proper attribution in academic contexts. While many universities are still operating in a Detect-React-Prevent Response (Jolley and Maimone, 2022), despite acknowledging the limitations of Gen-AI detection software, there is an increasing need to move towards looking at Gen-AI as a resource and using the Integrate-Educate-Model Approach (Jolley and Maimone, 2022). This approach encourages the integration of Gen-AI into the curriculum and highlights best practices, therefore upskilling staff and students (Wise et al., 2024).

In addition to ethical concerns, questions about access to Gen-AI technologies have garnered significant attention, which highlights the risk of deepening inequalities if Gen-AI is not made accessible across diverse socio-economic contexts (Langa et al., 2025; Langeveldt and Pietersen, 2024; Maimela and Mbonde, 2025; UNESCO, 2023). These perspectives align with global initiatives aimed at ensuring that AI technologies benefit all stakeholders fairly, in particular in academic and research settings. Higher education institutions would require institutional IT capacity to ensure equitable access by the adoption of vetted Gen-AI technologies that would also ensure academic integrity, data privacy, and informed consent.

The integration of Gen-AI into academia offers transformative potential but also presents complex challenges. Concerns about transparency and stigma, coupled with broader ethical and access-related considerations, underscore the need for carefully calibrated policies. Frameworks must navigate the fine line between encouraging innovation and safeguarding academic integrity and institutional data security. This will ensure that Gen-AI tools are adopted responsibly and inclusively.

1.2 South Africa's National AI Policy Framework

South Africa's *National Artificial Intelligence Policy Framework* is a forward-thinking approach to addressing the ethical and societal implications of AI adoption (Department of Communications and Digital Technologies, 2024). The framework emphasises ethical AI development,

transparency, and fairness while recognising the importance of inclusivity and capacity-building. A key focus is promoting human-centred AI that complements rather than replaces human decision-making. The framework also seeks to bridge the digital AI divide by integrating Gen-AI into educational curricula and fostering public-private partnerships to enhance capacity and innovation. By prioritising these efforts, the framework aligns with global standards while addressing South Africa's unique socio-economic challenges. Importantly, it advocates transparency in Gen-AI use while recognising the need to avoid creating barriers to its responsible adoption.

In conclusion, South Africa's *National AI Policy Framework* is a holistic strategy for integrating Gen-AI into academia and beyond. By addressing both local and global challenges it provides a blueprint for fostering innovation while ensuring ethical governance and equitable access.

1.3 Scope and structure of this article

The UKZN *AI Academic Guidelines* identify three interrelated objectives for Gen-AI adoption in academic contexts: *augmenting academic productivity*, *stimulating innovation*, and *facilitating interdisciplinary collaboration*. This article focuses on those aspects of the Guidelines most directly engaged in current debates on academic integrity, ethical AI use, and the practical enablers of institutional capacity-building. While all three objectives are interconnected, the analysis emphasises productivity and innovation because these are most contested in scholarly and policy discourse, and because they intersect most directly with the pedagogical and ethical principles set out in the Guidelines. Interdisciplinary collaboration, while central to the Guidelines' vision, is addressed more implicitly, primarily as an underlying value in the collaborative process through which the Guidelines were developed and in their anticipated role in fostering cross-disciplinary engagement with Gen-AI.

The article proceeds as follows: Section 2 sets out the qualitative, single-institution case study design, combining process tracing and comparative policy analysis, and documents the eight-stage pathway that produced the Guidelines. Section 3 presents the discussion, structured around seven aspects of Gen-AI use in academia that we suggest are most critical. The article then concludes by distilling implications for African higher education and outlining priorities for iterative refinement of the Guidelines.

2 Methodology

2.1 The UKZN AI Academic Guidelines as a case study

This study adopts a qualitative, single-institution case study approach to document and critically reflect on the development of the UKZN *AI Academic Guidelines* (Amin et al., 2025). The case study focuses on the full cycle of guideline development, from inception to final institutional approval, drawing directly on the authors' experience as members of UKZN's AI Task Team, supplemented by detailed records of the team's work, supporting documents, and contemporaneous meeting notes. The research aim is to examine how

an African public research university operationalised national AI policy priorities within an enabling institutional framework. This article has three interrelated objectives:

1. *To document the process* by which the UKZN AI Academic Guidelines were developed, from initial conception through to formal institutional adoption.
2. *To analyse* how normative principles were operationalised in the final Guidelines through specific provisions and implementation mechanisms.
3. *To distil transferable insights* for other higher education institutions seeking to align their own policies with national AI priorities.

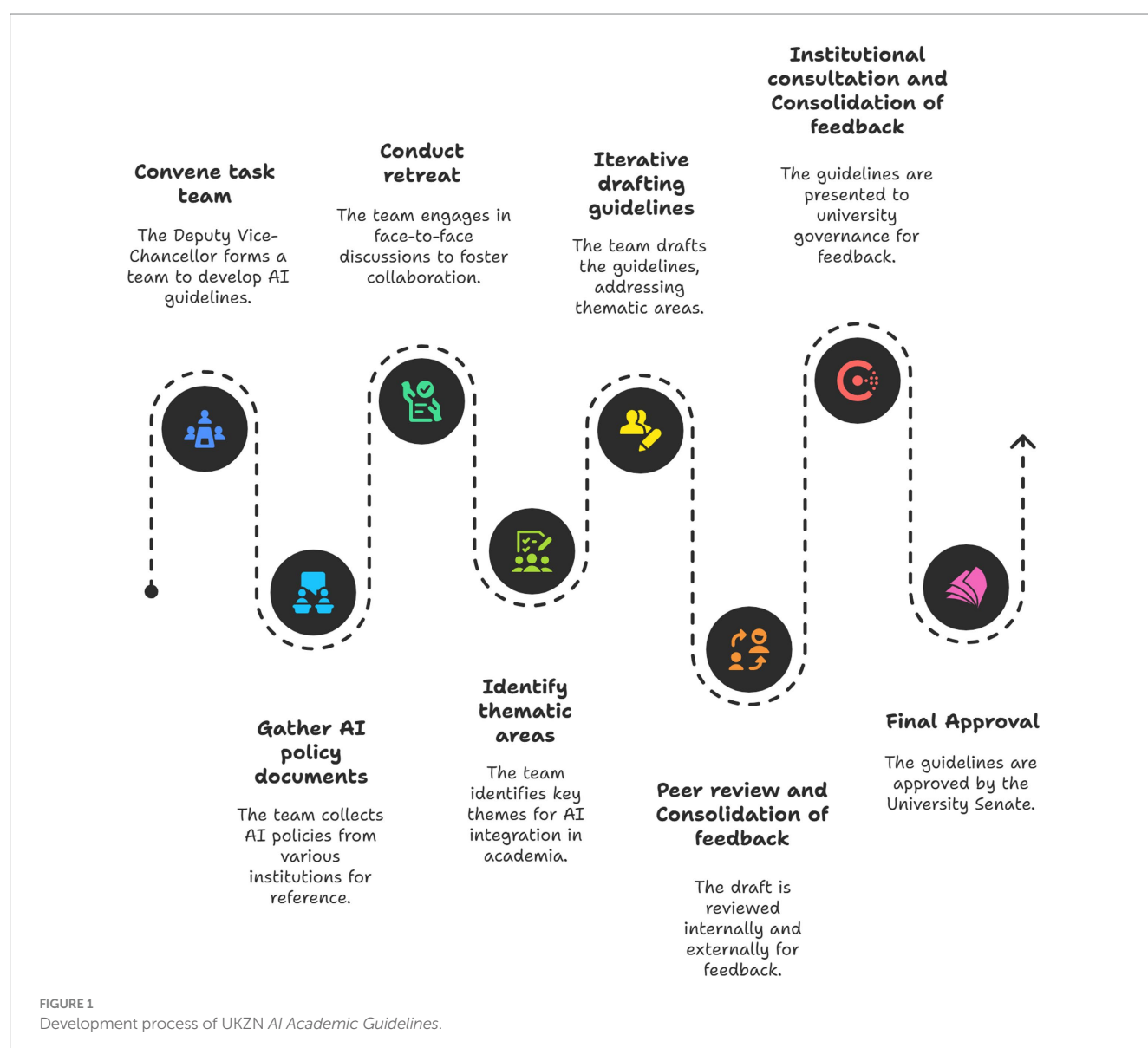
This Methodology section focuses primarily on documenting the development process (Objective 1). The subsequent Discussion section addresses the remaining objectives by analysing the Guidelines' operational provisions (Objective 2) and distilling lessons for other institutions (Objective 3).

2.2 The process of developing the UKZN AI Academic Guidelines

The development of UKZN's AI Academic Guidelines was a structured, consultative, and interdisciplinary process. Figure 1 depicts the eight key stages of the development.

In mid-2024, the Deputy Vice-Chancellor for Teaching and Learning convened a task team comprising approximately a dozen members drawn from various disciplines and Colleges across the university. The team's mandate was to formulate institutional guidelines for the appropriate integration of Gen-AI into academic practice.

To foster collaboration and enable in-depth deliberation on potentially contentious issues, the task team commenced its work with an in-person retreat. The primary aim of the retreat was two-fold: first, to cultivate collegiality and mutual understanding among members and, second, to engage in sustained, face-to-face debate with the resolution of key issues. This approach was considered more effective for complex normative deliberations than asynchronous exchanges or online meetings.



In preparation for the retreat, and with the support of the Office of the DVC, the task team gathered a wide array of AI policy documents from South African and international institutions. These were then shared among members and served as a comparative resource during initial discussions. During the retreat, the task team identified thematic areas central to the use of Gen-AI in academia. Each theme was deliberated upon in depth, with the aim of reaching principled consensus and ensuring internal coherence across the prospective guidelines. Once agreement had been reached in principle, the members divided into smaller drafting groups, each of which was assigned a thematic area to translate into practical policy text.

As expected, the drafting process raised further questions, which were brought back to the plenary for additional discussion and resolution. Following several iterative cycles of drafting and deliberation, the task team then consolidated the work into a final draft of the Guidelines. This draft was then submitted for peer review, both internally in the university and externally to two international experts. The purpose of including external reviewers was to obtain a fresh, independent perspective that was uncoloured by the team's internal deliberations. After incorporating the feedback from both internal and external reviewers, the task team commenced a formal institutional consultation process. During this phase, the chair of the task team presented the draft Guidelines to all relevant university governance structures. The Guidelines were circulated in advance of each presentation, and meetings entailed open discussion, feedback, and clarifications, often with the support of other task team members. The Guidelines were not presented as a *fait accompli* but rather as a well-reasoned proposal that was open to constructive refinement. All feedback—whether questions, concerns, or suggestions—was formally considered and then responded to in detail.

Following this comprehensive consultative process, and after incorporating further revisions based on institutional feedback, the final draft of the UKZN *AI Academic Guidelines* was submitted to the University Senate, where it was formally approved.

The resulting UKZN *AI Academic Guidelines* reflect a process grounded in broad-based participation, comparative policy analysis, and iterative consensus-building. The following discussion builds on this methodological foundation by analysing how the Guidelines operationalise their stated aims. In doing so, it links the process of their creation to the substantive provisions that emerged.

3 Discussion

The UKZN *AI Academic Guidelines* are anchored in four foundational principles—encouraging innovation, ethical and responsible use, academic rigour and integrity, and capacity-building—which reflect the university's commitment to fostering innovation, maintaining academic integrity, and preparing its community for the transformative potential of Gen-AI. These principles serve as the normative lens through which the Guidelines address the practical realities of AI integration in higher education.

For the purposes of this article, we focus on those aspects of the Guidelines that best illustrate how these foundational principles are operationalised in practice, organised into seven subsections: (3.1) the institutional stance of encouraging responsible Gen-AI use; (3.2) academic rigour and human oversight in AI-assisted work; (3.3) integration of Gen-AI literacy into curricula with module-level flexibility; (3.4) AI-generated text and plagiarism; (3.5) disclosure in

academic research publications; (3.6) implementation and capacity-building; and (3.7) monitoring, evaluation, and continuous improvement. These seven aspects were selected because they present the most significant opportunities and challenges for integrating Gen-AI into academic life, and because they capture the breadth of the Guidelines' pedagogical, ethical, and operational provisions.

3.1 Encouraging the use of Gen-AI

The UKZN *AI Academic Guidelines* take a proactive stance in encouraging the use of Gen-AI tools in academic work, provided their application adheres to the principles outlined in the Guidelines. This approach recognises the potential of Gen-AI technologies to enhance and augment academic productivity by automating and streamlining laborious routine tasks such as reviewing the literature, drafting and editing content, synthesising complex data, and facilitating innovative research design. By enabling researchers and students to focus on higher-order intellectual pursuits, Gen-AI serves as a valuable complement to traditional academic skills, thereby fostering critical higher-order thinking, innovative intellectual exploration, greater creativity, and interdisciplinary collaboration (Butson and Spronken-Smith, 2024). The Guidelines serve as the basis to guide academics on whether and how to integrate Gen-AI in the classroom, and to allow students to develop the skills and knowledge required to succeed in a technologically advancing world. This institutional leadership is particularly significant given that Africa as a whole lags in AI governance compared to the Global North, although South Africa has shown comparatively greater progress (Cloete, 2024). Recent meta-synthesis evidence further confirms that most African higher education institutions remain in early or aspirational stages of AI governance, with only a handful—such as those in South Africa, Rwanda, and Nigeria—beginning to align institutional policy with national AI strategies (Sangwa et al., 2025; Sebihi et al., 2025).

This proactive stance can be further understood through the lens of Constructivist Learning Theory, which emphasises that learners build knowledge actively rather than passively absorbing information (Grubaugh et al., 2023). In this framework, Gen-AI functions as a “cognitive partner” that supports exploration, hypothesis testing, and creative problem-solving. Scholarship has argued that Gen-AI can act as a “more knowledgeable other” within a social constructivist framework, scaffolding learning in the zone of proximal development and contributing to the co-construction of knowledge between humans and AI (Tran et al., 2025, p. 1). Complementary perspectives reinforce this view: ChatGPT has been framed through a socio-technical lens as augmenting rather than replacing human cognition in South African universities (Tarisayi, 2024); and AI has been described as a disorienting dilemma that prompts critical reflection and adaptation in assessment practices, consistent with Transformative Learning Theory (Ragolane and Patel, 2024). Empirical studies using activity theory further demonstrate that conversational AI tools already mediate academic practices by supporting teaching, research, and administration while simultaneously reshaping traditional methods (Venter et al., 2025); and South African lecturers report that AI can both enhance and transform pedagogy, consistent with the SAMR model of educational technology integration, which categorises technology use into four stages: substitution, augmentation, modification, and redefinition (Sanders and Mukhari, 2024). Finally, systematic review evidence confirms these benefits, finding that

ChatGPT enhances student engagement, personalised feedback, and conceptual understanding in South African universities, while also highlighting risks of plagiarism and over-reliance (Sokhulu et al., 2025). By framing AI as a tool for meaning-making rather than a shortcut to answers, the Guidelines promote learner autonomy, foster critical engagement with information, and encourage students to integrate AI outputs into their own reasoning processes.

Encouraging Gen-AI use is also aligned with South Africa's broader socio-economic development goals (Opesemowo and Adekomaya, 2024). By equipping students and academics with AI literacy, the Guidelines support the development of a skilled workforce that is prepared for the demands of an increasingly technology-driven global economy. This emphasis positions UKZN as a leader, both locally and internationally, in adopting AI responsibly and ensures that its graduates remain competitive while contributing to national priorities, such as bridging the digital AI divide and promoting inclusive innovation. Gen-AI, therefore, becomes not just a tool for individual efficiency but also a driver of societal progress.

While much of the current discourse—and indeed much of this paper—focuses on the use of Gen-AI in text-based academic work, the UKZN *AI Academic Guidelines* are equally applicable to a broader range of AI-supported academic activities. These include image generation for visualisation, preliminary data analysis, simulation, and quantitative modelling, as reflected in the Guidelines' provisions for academic activities such as coding, statistical interpretation, and pedagogical content creation (Amin et al., 2025, p. 10). Such applications, while generally attracting fewer concerns about plagiarism or authorship, still require human oversight to ensure accuracy, methodological soundness, and ethical use. Acknowledging these domains highlights that Gen-AI integration in academia extends beyond the contested terrain of text generation, encompassing a wide spectrum of tools that can enhance research, teaching, and learning.

The endorsement of Gen-AI use by the Guidelines is coupled with safeguards to ensure its responsible integration. While Gen-AI is encouraged for its ability to enhance productivity and innovation, the Guidelines stress the importance of transparency, sound methodology, and human oversight (Amin et al., 2025, p. 3). This ensures that Gen-AI tools are used ethically and in ways that preserve academic integrity. By providing a structured framework for the adoption of Gen-AI, UKZN fosters a culture of innovation that is both ambitious and grounded in the core values of academia.

In developing these Guidelines, we evaluated several alternative approaches—limiting Gen-AI strictly to tasks like proofreading, imposing a blanket ban on AI, or requiring full disclosure for every AI interaction—and found each option to be ultimately lacking. Restricting Gen-AI to narrow functions would be difficult to monitor and would limit genuine innovation. Banning AI outright would be a knee-jerk response that would deprive students and staff of essential skills in a competitive global landscape. We were deeply concerned about Gen-AI hollowing out student learning, and that requiring detailed disclosure for every use of AI would create unnecessary administrative burdens and hinder normal academic practice. Accordingly, we concluded that encouraging Gen-AI with safeguards, rather than confining, prohibiting, or over-regulating it, most effectively supports our commitment to academic integrity, innovation, and broader socio-economic development.

3.2 Academic rigour in the context of AI use

The UKZN *AI Academic Guidelines* emphasise the importance of maintaining academic rigour in the context of Gen-AI use. Rigour, in this context, involves validating the accuracy and reliability of AI-generated outputs, and ensuring that they meet the same scholarly standards as traditionally produced work (Amin et al., 2025, pp. 6–8). This includes scrutiny of AI-generated text, ideas, or data to verify their alignment with sound research methodologies. The Guidelines underscore that while Gen-AI tools can assist in academic work, they must not replace critical human judgement, which remains central to maintaining the credibility of scholarly outputs (Amin et al., 2025, p. 6).

This commitment to rigour is further illuminated by Self-Regulated Learning (SRL) Theory, which identifies three core phases of effective learning: forethought (goal setting and strategic planning), performance (self-monitoring and strategy use), and self-reflection (evaluating and adapting approaches) (Brenner, 2022). When students use Gen-AI tools within this cycle, they must set clear academic objectives, critically monitor AI-generated outputs for accuracy and relevance, and reflect on how those outputs contribute to their own learning goals. This operationalises rigour as an active, student-driven process rather than a static standard imposed externally.

Central to this rigour is the importance of engaging with primary sources, which remains a critical aspect of credible academic work (Amin et al., 2025, p. 8). This necessity is underscored by examples from the pre-AI era, where reliance on secondary sources often led to the perpetuation of inaccuracies and thereby weakened academic foundations. For instance, cases where scholars did not verify claims against original texts highlight the risks of bypassing primary materials, thus undermining the reliability of their work. The Guidelines extend this lesson to the use of Gen-AI, and stress that while AI tools can synthesise secondary sources or generate drafts, they must not replace direct engagement with the primary evidence. By maintaining this standard, the Guidelines ensure that Gen-AI serves as a tool to enhance scholarly inquiry without compromising its foundational integrity.

The Guidelines also stress the need for students and researchers to engage critically with AI-generated content to prevent over-reliance on these tools (Amin et al., 2025, p. 5). This includes understanding the limitations of AI, such as potential biases or inaccuracies, and actively addressing these shortcomings in academic work. By embedding academic rigour into AI use, the Guidelines ensure that Gen-AI tools enhance rather than diminish the quality and reliability of academic outputs, thereby safeguarding the core values of scholarship in an evolving technological landscape.

It is essential that both staff and students receive structured guidance in the development of effective prompts, i.e., prompt 'engineering' or 'crafting', when engaging with Gen-AI tools. While such tools may appear straightforward to use, their outputs are significantly influenced by the quality and clarity of the prompts provided (Lin, 2024, 2023). Effective prompting involves crafting inputs that provide sufficient relevant context, assign specific roles, and include key information to direct the generative process (Lin, 2023). This often requires iterative refinement to ensure that the output is relevant, accurate, and aligned with the intended academic

purpose. In parallel, users must also be trained to critically assess the generated content in terms of its factual accuracy, contextual appropriateness, and overall usefulness in an academic framework. Without such evaluative skills, there is a heightened risk of uncritical acceptance of content that may be misleading, superficial, or inappropriate for scholarly use (Lin, 2024, 2023).

In formulating this approach, we weighed and ultimately dismissed several alternative methods for preserving academic rigour: for instance, confining Gen-AI use strictly to minor tasks, like grammar checks. This however would limit its potential to accelerate complex data analysis and research design, whereas granting AI near-autonomous control over inquiry would risk eroding human-driven critical judgement. We also considered imposing mandated Gen-AI cross-checks of all sources and claims but concluded that such a universal requirement could stifle the discipline-specific flexibility that is essential for robust scholarship. Consequently, the chosen path strikes a balance—insisting on direct engagement with primary evidence, proper scrutiny of AI-generated content, and a clear acknowledgement of AI's limitations—therefore ensuring that Gen-AI remains a tool to supplement, and not supplant, the foundations of academic excellence.

3.3 Integration of Gen-AI literacy into curricula: flexibility at module-level

UKZN's commitment to integrating Gen-AI literacy and skills into its curricula aligns with the principles of South Africa's *National Artificial Intelligence Policy Framework*, which emphasises talent development and the integration of AI into education in order to foster innovation and inclusivity (Department of Communications and Digital Technologies, 2024). By actively encouraging the use of Gen-AI and embedding it into academic programmes, UKZN ensures that students and staff are equipped to navigate an AI-driven future. This approach complements traditional academic skills (such as critical thinking, problem-solving, and data analysis), thereby positioning Gen-AI as a tool to enhance intellectual capabilities. The Guidelines align with the national vision of leveraging AI to drive socio-economic development while preparing a workforce that is capable of harnessing the opportunities that AI presents. To translate this strategic vision into effective pedagogy, the integration of Gen-AI is framed through established learning theory, ensuring that its adoption strengthens—not supplants—core educational outcomes.

The integration of Gen-AI literacy into curricula can be conceptualised using Bloom's Digital Taxonomy (Churches, 2008), which updates Bloom's classic cognitive hierarchy for digital learning contexts. Gen-AI tools can support foundational learning goals such as remembering and understanding, but their transformative potential lies in higher-order skills: analysing AI outputs for bias or error, evaluating the quality of AI-assisted work, and creating new, original artefacts that synthesise human insight with AI capabilities. By embedding these higher-order applications into curricula, the Guidelines ensure that Gen-AI use develops advanced academic competencies rather than replacing them.

The Guidelines take a flexible approach to the integration of Gen-AI into teaching and learning (Amin et al., 2025, pp. 9–16). While the use of Gen-AI tools is considered the default, individual

modules may impose limits on AI use when this is justified by sound pedagogical reasons. This flexibility ensures that Gen-AI tools are used thoughtfully, thereby aligning with specific learning objectives while preserving academic autonomy in the university. Such an approach recognises the diversity of academic disciplines and their varying needs. Faculties are encouraged to integrate Gen-AI into their courses, have proactive discussions with students on Gen-AI use and its impact on their learning, use AI as a learning partner, ethically use AI, and have clear class policies on AI use for graded tasks. Concomitant with the increased use of and access to Gen-AI tools is the need for academics to critically re-evaluate assessment practices so that critical thinking and problem-solving skills are assessed rather than rote-learning of AI-generated essays/assignments.

In deciding how best to integrate Gen-AI literacy into teaching and learning, and in keeping with South Africa's *National Artificial Intelligence Policy Framework* (Department of Communications and Digital Technologies, 2024), we quickly dismissed two extremes: banning Gen-AI in all foundational modules or making AI integration purely optional. A total ban would ensure that we get genuine student engagement but would impede innovation and clash with the national push for inclusive AI uptake, while making Gen-AI optional could leave many students unprepared for a world increasingly dominated by digital processes. Instead, we chose a middle path: Gen-AI is treated as the default, but individual modules can limit its use when that is clearly justified by pedagogical needs. This approach fosters broad AI readiness while preserving the flexibility to adapt across diverse academic disciplines.

3.4 AI-generated text and plagiarism

The UKZN *AI Academic Guidelines* address a prevalent misconception that the use of AI-generated text automatically constitutes plagiarism, by making it clear that it does not. Several factors underpin this reasoning. First, Gen-AI lacks moral agency, meaning it cannot independently claim authorship or responsibility for its outputs (Amin et al., 2025, p. 8). Editorial policies across leading international journals have converged on the same principle, prohibiting AI from being credited as an author while requiring disclosure of its use and reinforcing that accountability rests with human researchers (Yoo, 2025). Gen-AI operates purely as a tool that involves executing tasks based on algorithms and human-provided prompts without any capacity for intellectual creativity or ethical judgement. Second, the academic using AI plays a pivotal role in directing and refining the generated content. This process involves crafting specific prompts, interpreting the outputs, and making substantial modifications, akin to a director orchestrating a performance where the vision and execution rest firmly with the human. Finally, AI-generated outputs are rarely used in isolation in academic work. Scholars critically evaluate, transform, and integrate these outputs with their own insights and analysis, a process that requires significant intellectual effort.

The Guidelines, however, caution that presenting ideas or insights generated by AI as one's own risks *idea plagiarism*, mainly because such ideas could have existed in the literature (Amin et al., 2025, p. 9). Gen-AI tools often synthesise information from a vast corpus of existing knowledge without attributing specific sources.

While the outputs may appear original, they frequently reflect ideas, arguments, or findings that already exist in the academic or broader intellectual discourse. In this regard, Hutson highlights the blurred boundaries between human-authored and AI-assisted content, questions traditional definitions of originality and intellectual property and emphasises the potential for both misuse and enhancement of educational outcomes (Hutson, 2024). Baron, however, draws attention to the growing inadequacy of traditional plagiarism detection methods in the face of increasingly sophisticated Gen-AI models like ChatGPT (Baron, 2024). Importantly, his argument is not that AI-generated content escapes being classified as plagiarism merely because it is difficult to detect. Rather, Baron's point is that an over-reliance on detection tools risks missing the deeper challenge: the urgent need to re-evaluate and redesign assessment strategies to preserve academic integrity in a transformed technological environment. By situating his concerns in the practical domain, Baron reinforces the principle that academic integrity must be maintained through thoughtful pedagogical responses—and not be abandoned because enforcement has become more difficult.

Suppose that AI-generated ideas are incorporated into scholarly work without acknowledgement of the original work. In that case, there is a risk of inadvertently claiming credit for concepts that are not genuinely novel, thereby violating principles of academic integrity. Scholars have a responsibility to ensure that AI-generated insights are cross-checked against primary sources and the broader body of knowledge to determine whether they represent genuinely novel contributions or are merely articulations of existing ideas. By neglecting this critical step, researchers risk not only perpetuating existing work without proper citation but also undermining the credibility of their own contributions. The Guidelines therefore emphasise that Gen-AI tools should be used as aids in the research process, and not as substitutes for the rigour required to produce original scholarship.

Building on this principle, the UKZN *AI Academic Guidelines* explicitly assign responsibility for verifying the originality and attribution of ideas to the individual user. Paragraph 6.1.3 requires that “when using ideas suggested by generative AI, it is essential to be familiar with the relevant literature to properly attribute existing concepts” (Amin et al., 2025, p. 8), while paragraph 6.2.2 obliges users to “conduct their own thorough literature searches to ensure that all ideas presented, even those generated by AI, are properly credited to their original sources” (Amin et al., 2025, p. 8). In practice, this means that any idea generated by a Gen-AI tool should be treated as potentially pre-existing, and its novelty tested through an independent literature review before inclusion in scholarly work. While this responsibility rests squarely on the human scholar, emerging Gen-AI features—such as citation traceability or side-by-side source text displays—can help streamline the process. These tools, however, can only complement, not replace, the critical oversight envisaged in the Guidelines.

The role of Gen-AI in academic work exists on a spectrum, from offering supplementary support that enhances efficiency to playing a significant role in generating substantive outputs. While some seek a clear-cut distinction between AI complementing and supplanting human ingenuity, in practice the boundary is often

fluid. The Guidelines address this ambiguity by requiring that, in all cases, the human academic retains creative direction, critical oversight, and final responsibility for the content produced. This ensures that, regardless of AI's degree of involvement, academic integrity and intellectual accountability remain grounded in human agency.

In shaping these Guidelines on plagiarism and AI-generated text, we weighed several alternatives: an outright ban on AI-produced text would eliminate ambiguity but limit valuable and increasingly improved academic tools. Furthermore, mandating comprehensive citation for all AI-assisted outputs—even minor grammatical suggestions—would create excessive administrative burdens. Ultimately, we opted to clarify that AI-generated text is not, by default, plagiarism, but cautioned scholars against presenting unacknowledged AI-derived ideas as being original. This choice preserves the benefits of Gen-AI whilst protecting academic integrity, so striking a balance between total prohibition and overly exhaustive disclosure requirements.

3.5 Disclosure in academic research publications

The UKZN *AI Academic Guidelines* outline a thoughtful and nuanced approach to disclosing the use of Gen-AI in academic research and related activities. This tiered system balances the need for transparency and accountability with the normalisation of Gen-AI use, thereby ensuring that its role in scholarly work is neither overemphasised nor unfairly stigmatised. By avoiding the risks of over-disclosure, such as stigma or heightened scrutiny (BaHammam, 2023), the Guidelines enable seamless integration of Gen-AI into academic practices while upholding ethical standards.

3.5.1 Scope of disclosure

The Guidelines apply to formal academic work by academics and students that contributes to advancing knowledge, including research publications, academic books, and peer-reviewed reports. Informal academic outputs, such as editorials, blogs, and public reports, are exempt from disclosure requirements, as their focus is primarily communication rather than being research outputs (Amin et al., 2025, pp. 16–17). This distinction ensures that disclosure requirements are applied judiciously, reserving detailed transparency for outputs that directly influence scholarly disciplines.

3.5.2 Levels of disclosure

The Guidelines establish three levels of disclosure that are tailored to the extent of Gen-AI involvement (Amin et al., 2025, pp. 17–18):

1. *Optional disclosure*: AI use that does not significantly contribute to intellectual content, such as language improvement or compiling a reference list, does not require acknowledgement. These uses are likened to human editorial assistance, which typically goes uncredited.
2. *Disclosure without prompts*: For AI applications that significantly contribute to intellectual content (e.g., generating ideas or drafting sections) but do not affect research methodology, their

use should be disclosed. However, the specific prompts or inputs provided to the AI used need not be detailed.

3. *Full disclosure*: When AI tools directly influence the validity or reliability of research, such as analysing data, shaping hypotheses, or developing predictive models, both the use of AI and the specific prompts or materials provided must be disclosed. This ensures that the AI's role in methodological aspects is transparent and open to evaluation.

3.5.3 Practical guidance on disclosure

The Guidelines provide clear instructions on where and how Gen-AI use should be disclosed (Amin et al., 2025, pp. 18–19). Methodological contributions by Gen-AI are to be detailed in the methodology section or in supplementary files, thereby ensuring clarity on AI's role in shaping research outcomes. For less impactful uses, such as drafting or idea generation, disclosure can be made in the acknowledgement section. Researchers must also detail the AI tool's name, version, and purpose of use but are not required to specify the dates of use.

3.5.4 Ethical considerations

The Guidelines categorically state that Gen-AI tools cannot be credited as co-authors, as they lack the capacity for intellectual responsibility and accountability, which are essential criteria for authorship (Amin et al., 2025, p. 18). Similarly, Gen-AI tools cannot be cited as sources. If Gen-AI provides information or ideas, it is the researcher's responsibility to trace and cite the original sources. This policy reinforces the importance of human oversight and accountability in all stages of research.

In addition to the disclosure requirements, the UKZN *AI Academic Guidelines* incorporate specific provisions on privacy-preserving AI use. Paragraph 6.10 ("When to use secure generative AI tools") requires staff and students to use secure AI systems—operating offline or otherwise configured to avoid external data sharing—when processing sensitive personal information or protectable intellectual property (Amin et al., 2025, p. 20). Paragraph 7.9 ("Institutional guidance of secure generative AI") mandates that the University's IT Services, in consultation with the relevant DVCs and the AI Thought Leadership Forum, maintain and publish a list of such secure tools (Amin et al., 2025, p. 23). As part of implementing these provisions, UKZN is prioritising an institutional licensing arrangement with a Gen-AI provider. This will ensure that any personal information included in prompts or data inputs is processed solely for the intended purpose, with robust security safeguards, thereby enabling compliance with the Protection of Personal Information Act (POPIA) while supporting responsible, large-scale adoption of generative AI across the University.

By structuring disclosure requirements around the nature and significance of AI use, the Guidelines encourage the responsible integration of Gen-AI into academic work. This tiered approach ensures that Gen-AI's contributions are recognised where relevant, while preserving the integrity and transparency of scholarly research. It strikes an essential balance between normalising Gen-AI use and maintaining rigorous academic standards.

In formulating this three-tiered system of disclosure, we examined alternative models that ranged from enforcing full reporting of every Gen-AI interaction, including minor uses like grammar checks, to having no disclosure requirements at all, which could obscure AI's influence on scholarly work. A blanket mandate to list every prompt

was deemed excessively burdensome for both authors and reviewers, while the absence of disclosure risked undermining accountability and academic integrity. By adopting a structured and yet flexible approach, we balance practical concerns with the need to identify where Gen-AI meaningfully shapes intellectual content, thereby encouraging both transparency and normalisation of AI's role in academic research without imposing unnecessary barriers.

3.6 Implementation and capacity-building

The UKZN *AI Academic Guidelines* recognise AI literacy as a cornerstone of responsible adoption. Paragraph 7.1 commits the University to "promote institution-wide AI literacy among students and staff, ensuring that all members of the academic community are equipped to use AI tools responsibly, critically, and effectively" (Amin et al., 2025, p. 20). This commitment is framed as an ongoing process that adapts to the evolving capabilities and risks of Gen-AI technologies.

To ensure consistent and effective roll-out, implementation responsibilities are embedded within UKZN's existing governance structures rather than creating new bureaucratic layers. The College Deans of Research and the College Deans of Teaching and Learning carry primary responsibility for overseeing the Guidelines' integration at College level, reporting to the Deputy Vice-Chancellors for Teaching and Learning and for Research. This arrangement leverages established institutional lines of accountability and ensures that AI-related initiatives remain aligned with broader university priorities.

Capacity-building is central to successful implementation. UKZN's AI Task Team is currently developing practical resources, including an AI literacy training curriculum, a model AI integration plan at School level, and a detailed online manual. By providing practical tools for adoption, UKZN aims to translate the Guidelines' principles into sustainable practice and to offer a model that other African universities can adapt.

3.7 Monitoring, evaluation, and continuous improvement

A core consideration in ensuring that the UKZN *AI Academic Guidelines* remain practical and fit for purpose is the establishment of a framework for ongoing monitoring and evaluation. Scholarship has emphasised that, given the fast-evolving nature of generative AI in academia, such frameworks must be adaptive, participatory, and responsive to emerging challenges (Jin et al., 2025), while also stressing that effective governance requires policies that foster accountable assimilation rather than reactionary resistance (Tarisayi, 2024).

The centrepiece of this framework will be the AI Thought Leadership Forum—a cross-university platform for dialogue, reflection, and innovation in AI use for teaching, learning, and research (Amin et al., 2025, p. 21). The Forum will convene academic leaders, technical experts, and external stakeholders to:

- Share emerging practices and innovations.
- Identify operational and pedagogical challenges.
- Recommend periodic updates to the Guidelines to ensure ongoing relevance.

This approach extends monitoring beyond compliance, fostering a culture of critical engagement with AI across the university community. While substantive self-reflection on implementation outcomes will only be possible after a period of sustained application, the Thought Leadership Forum provides the mechanism for capturing feedback, promoting best practices, and ensuring that the Guidelines evolve in step with technological and educational developments.

4 Conclusion

The UKZN *AI Academic Guidelines* represent a pioneering institutional effort from the African continent to develop a responsible, enabling framework for integrating Gen-AI into higher education. In contrast to restrictive or purely compliance-driven models, the Guidelines promote an approach that encourages innovation while safeguarding academic integrity. By addressing critical issues such as transparency, plagiarism, and capacity-building, they provide a practical and ethically grounded pathway for adopting Gen-AI across diverse academic contexts.

Importantly, these Guidelines emerge from a Global South setting—characterised by resource constraints, infrastructural diversity, and a strong imperative to bridge digital divides (Maimela and Mbonde, 2025; Sebihi et al., 2025). As such, they offer a valuable and underrepresented perspective in the international conversation on educational AI policy. The UKZN experience shows how institutions in Africa can actively shape trustworthy and inclusive AI integration strategies, grounded in local realities but informed by global standards.

By sharing our institutional approach and underlying reasoning, we aim to contribute meaningfully to global discussions on the governance of AI in higher education. We invite collaboration from scholars, educators, and policymakers worldwide, particularly those working in low- and middle-income contexts, to ensure that future-facing AI strategies are both inclusive and adaptable.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material further inquiries can be directed to the corresponding author.

Author contributions

DT: Writing – original draft, Writing – review & editing. MB: Writing – original draft, Writing – review & editing. AB: Writing – original draft, Writing – review & editing. HC: Writing – original draft, Writing – review & editing. SDu: Writing – original draft,

Writing – review & editing. SDI: Writing – original draft, Writing – review & editing. NA: Writing – original draft, Writing – review & editing. WH: Writing – original draft, Writing – review & editing. RG: Writing – original draft, Writing – review & editing. JB-B: Writing – original draft, Writing – review & editing. AV: Writing – original draft, Writing – review & editing. NK: Writing – original draft, Writing – review & editing. AC: Writing – original draft, Writing – review & editing.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The authors declare that Gen-AI was used in the creation of this manuscript. The authors acknowledge the assistance of ChatGPT-4 in drafting the initial version of this manuscript and in enhancing its language and readability, and subsequently ChatGPT-5 in assisting with revisions. The authors have read and approved the final version of the manuscript and take full responsibility for its content.

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References

- Amin, N., Badru, A., Botes, M., Bruce-Band, J., Chenia, H., and Chuturgoon, A. A. (2025). Principles and guidelines on the use of generative artificial intelligence tools in academic work. Durban: University of KwaZulu-Natal.
- BaHammam, A. S. (2023). Balancing innovation and integrity: the role of AI in research and scientific writing. *Nat. Sci. Sleep* 15, 1153–1156. doi: 10.2147/NSS.S455765

- Balalle, H., and Pannilage, S. (2025). Reassessing academic integrity in the age of AI: a systematic literature review on AI and academic integrity. *Soc. Sci. Human. Open* 11:101299. doi: 10.1016/j.ssaho.2025.101299
- Baron, P. (2024). Are AI detection and plagiarism similarity scores worthwhile in the age of ChatGPT and other Generative AI? *Scholarship Teach. Learn. South* 8, 151–179. doi: 10.36615/sotls.v8i2.411
- Brenner, C. A. (2022). Self-regulated learning, self-determination theory and teacher candidates' development of competency-based teaching practices. *Smart Learn. Environ.* 9:3. doi: 10.1186/s40561-021-00184-5
- Butson, R., and Spronken-Smith, R. (2024). AI and its implications for research in higher education: a critical dialogue. *High. Educ. Res. Dev.* 43, 563–577. doi: 10.1080/07294360.2023.2280200
- Cain, W. (2023). AI emergence in education: exploring formative tensions across scholarly and popular discourse. *J. Interact. Learn. Res.* 34, 239–273. doi: 10.70725/74960Somikun
- Cheong, L., Guo, A., Lee, M., Liao, Z., Kadoma, K., Go, D., et al. (2025). Penalizing transparency? How AI disclosure and author demographics shape human and AI judgments about writing. CHIWORK 2025 workshop on generative AI disclosure, ownership, and accountability in co-creative domains.
- Churches, A. (2008). Bloom's digital taxonomy.
- Cloete, F. (2024). Governing artificial intelligence (AI) and other technologies in the digital era. *Administratio Publica* 32, 1–30. doi: 10.10520/ejc-adminpub_v32_n1_a3
- Department of Communications and Digital Technologies (2024). SA national AI policy framework. Pretoria: Government Printer.
- Eke, D. O. (2023). ChatGPT and the rise of generative AI: threat to academic integrity? *J. Responsib. Technol.* 13:100060. doi: 10.1016/j.jrt.2023.100060
- Grubaugh, S., Levitt, G., and Deever, D. (2023). Harnessing AI to power constructivist learning: an evolution in educational methodologies. *JETM* 13, 81–83. doi: 10.59652/jetm.v1i3.43
- Harvard College Students and Teaching Fellows in Creativity (Gen Ed 1067) (Spring 2023) (last) Atherton, D., Newman, S., and Esfahany, K. (2023). Proposed Harvard AI code of conduct. Cambridge, Massachusetts (USA).
- Hosseini, M., Resnik, D. B., and Holmes, K. (2023). The ethics of disclosing the use of artificial intelligence tools in writing scholarly manuscripts. *Res. Ethics* 19, 449–465. doi: 10.1177/17470161231180449
- Hutson, J. (2024). Rethinking plagiarism in the era of generative AI. *J. Int. Commun.* 3, 20–31. doi: 10.54963/jic.v3i2.220
- Jin, Y., Yan, L., Echeverria, V., Gašević, D., and Martinez-Maldonado, R. (2025). Generative AI in higher education: a global perspective of institutional adoption policies and guidelines. *Comput. Educ. Artif. Intell.* 8:100348. doi: 10.1016/j.caeai.2024.100348
- Jolley, J. R., and Maimone, L. (2022). Thirty years of machine translation in language teaching and learning: a review of the literature. *L2 J.* 14, 26–44. doi: 10.5070/L214151760
- Khalifa, M., and Albadawy, M. (2024). Using artificial intelligence in academic writing and research: an essential productivity tool. *Comput. Methods Prog. Biomed. Update* 5:100145. doi: 10.1016/j.cmpbup.2024.100145
- Langa, V. G., Balkaran, S., Mahlala, S., and Muronda, B. (2025). Promoting social inclusion through AI integration in higher tertiary institutions: a South African perspective. *IJBES* 7, 299–308. doi: 10.36096/ijbes.v7i4.848
- Langeveldt, D. C., and Pietersen, D. (2024). Decolonising AI: a critical approach to education and social justice. *IJER* 6, 1–9. doi: 10.38140/ijer-2024.vol6.s1.07
- Lin, Z. (2023). How to write effective prompts for large language models. doi: 10.31234/osf.io/r78fc
- Lin, Z. (2024). Beyond principlism: practical strategies for ethical AI use in research practices. *AI Ethics* 5, 2719–2731. doi: 10.1007/s43681-024-00585-5
- Maimela, C., and Mbonde, P. (2025). Artificial intelligence in South African universities: curriculum transformation and decolonisation-aid or obstacle? *Front. Sociol.* 10:1543471. doi: 10.3389/fsoc.2025.1543471
- Maina, A. M., and Kuria, J. (2024). "Building an AI future: Research and policy directions for Africa's higher education" in 2024 IST-Africa conference (IST-Africa). Presented at the 2024 IST-Africa conference (IST-Africa) (Dublin, Ireland: IEEE), 01–09.
- Opesemowo, O. A. G., and Adekomaya, V. (2024). Harnessing artificial intelligence for advancing sustainable development goals in South Africa's higher education system: a qualitative study. *IJLTER* 23, 67–86. doi: 10.26803/ijlter.23.3.4
- Porsdam Mann, S., Vazirani, A. A., Aboy, M., Earp, B. D., Minssen, T., Cohen, I. G., et al. (2024). Guidelines for ethical use and acknowledgement of large language models in academic writing. *Nat. Mach. Intell.* 6, 1272–1274. doi: 10.1038/s42256-024-00922-7
- Ragolane, M., and Patel, S. (2024). Transforming educ-AI-tion in South Africa: can AI-driven grading transform the future of higher education? *JETM* 3, 26–51. doi: 10.58425/jetm.v3i1.267
- Rana, N. K. (2025). Generative AI and academic research: a review of the policies from selected HEIs. *Higher Educ. Future* 12, 97–113. doi: 10.1177/23476311241303800
- Reinmann, G. (2023). Deskillung durch Künstliche Intelligenz? Potenzielle Kompetenzverluste als Herausforderung für die Hochschuldidaktik. Berlin: Hochschulforum Digitalisierung.
- Resnik, D. B., and Hosseini, M. (2025). Disclosing artificial intelligence use in scientific research and publication: when should disclosure be mandatory, optional, or unnecessary? *Account. Res.*, 1–13. doi: 10.1080/08989621.2025.2481949
- Russell Group (2023). Russell Group principles on the use of generative AI tools in education. London: Russell Group Universities.
- Sanders, D. A., and Mukhari, S. S. (2024). Lecturers' perceptions of the influence of AI on a blended learning approach in a south African higher education institution. *Discov. Educ.* 3:135. doi: 10.1007/s44217-024-00235-2
- Sangwa, S., Ekosse, E., and Placide, M. (2025). AI governance in African higher education: Status, challenges, and a future-proof policy framework 2054. doi: 10.62617/aie2054
- Sebihi, A., Schoelen, L., and Uwamwezi, B. (2025). Unveiling AI's impact in Rwandan higher education: navigating trends, challenges, and imperatives at the University of Rwanda. *Discov. Educ.* 4:166. doi: 10.1007/s44217-025-00576-6
- Sokhulu, L., Zulu, M., and Lott-Naidoo, D. (2025). Artificial intelligence in higher education South Africa: the role of ChatGPT in students' learning. *J. Educ.* 142–166. doi: 10.17159/2520-9868/i99a08
- Sullivan, M., Kelly, A., and McLaughlan, P. (2023). ChatGPT in higher education: considerations for academic integrity and student learning. *JALT* 6, 31–40. doi: 10.37074/jalt.2023.6.1.17
- Tarisayi, K. S. (2024). ChatGPT use in universities in South Africa through a socio-technical lens. *Cogent Educ.* 11:2295654. doi: 10.1080/2331186X.2023.2295654
- Tran, M., Balasooriya, C., Semmler, C., and Rhee, J. (2025). Generative artificial intelligence: the 'more knowledgeable other' in a social constructivist framework of medical education. *npj Digit. Med.* 8:430. doi: 10.1038/s41746-025-01823-8
- UNESCO (2023). Harnessing the era of artificial intelligence in higher education: a primer for higher education stakeholders [WWW Document]. Available online at: <https://unesdoc.unesco.org/ark:/48223/pf0000386670> (Accessed June 23, 2025).
- Van Dis, E. A. M., Bollen, J., Zuidema, W., van Rooij, R., and Bockting, C. L. (2023). ChatGPT: five priorities for research. *Nature* 614, 224–226. doi: 10.1038/d41586-023-00288-7
- Venter, I. M., Blignaut, R. J., Cranfield, D. J., Tick, A., and Achi, S. E. (2025). AI versus tradition: shaping the future of higher education. *JARHE* 17, 151–167. doi: 10.1108/JARHE-12-2024-0702
- Wise, B., Emerson, L., Van Luyn, A., Dyson, B., Bjork, C., and Thomas, S. E. (2024). A scholarly dialogue: writing scholarship, authorship, academic integrity and the challenges of AI. *High. Educ. Res. Dev.* 43, 578–590. doi: 10.1080/07294360.2023.2280195
- Xiao, P., Chen, Y., and Bao, W. (2023). Waiting, banning, and embracing: an empirical analysis of adapting policies for generative AI in higher education. 1–33. doi: 10.48550/ARXIV.2305.18617
- Yoo, J.-H. (2025). Defining the boundaries of AI use in scientific writing: a comparative review of editorial policies. *J. Korean Med. Sci.* 40:e187. doi: 10.3346/jkms.2025.40.e187
- Yusuf, A., Pervin, N., and Román-González, M. (2024). Generative AI and the future of higher education: a threat to academic integrity or reformation? Evidence from multicultural perspectives. *Int. J. Educ. Technol. High. Educ.* 21:21. doi: 10.1186/s41239-024-00453-6